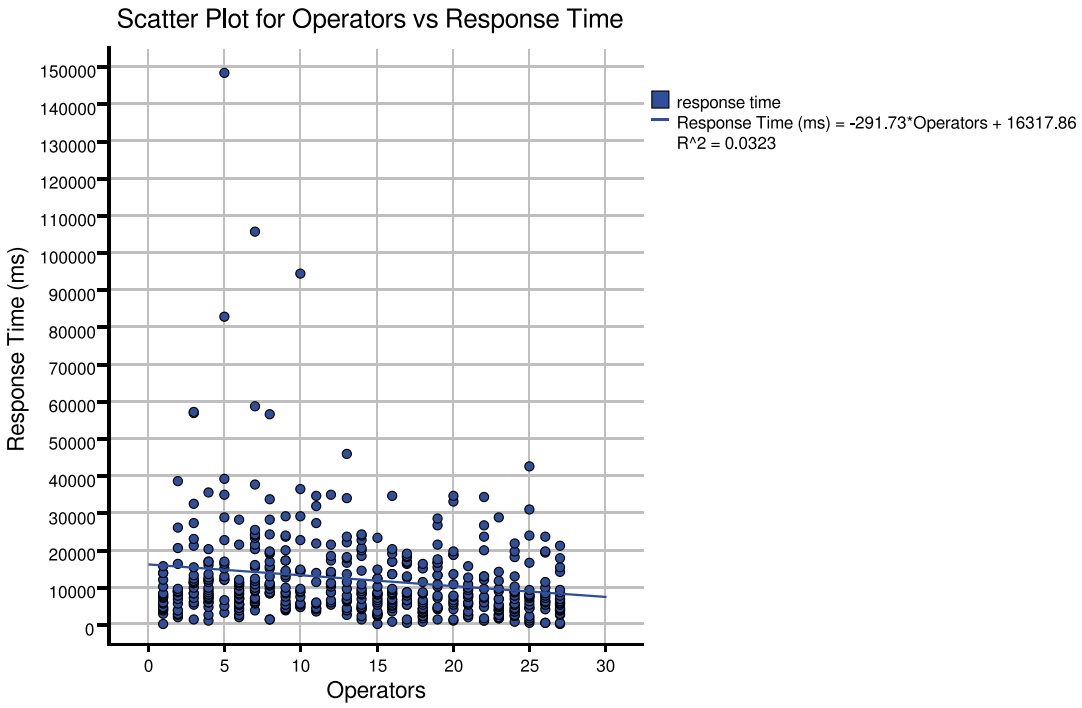


Fig. 9. Scatter Plot Operators vs Response Time

Regarding Hypothesis H2, the results of the One-Way ANOVA test and Tukey's HSD Multiple Comparison Test indicate that most operators do not show a significant difference in task completion time when compared to each other. A few operators, namely or, not, and left shift, appear to be significant. However, overall, the results support Hypothesis 2, suggesting that the complexity of bitwise operators does not result in longer task completion times.

### 4.2 Similarity of Results

While there are no directly comparable studies available for reference, it is important to contextualize our findings within the broader landscape of research on bitwise operators and programming language development. Our study contributes to this field by examining the intuitiveness of bitwise operators through task completion time and error rates, shedding light on their practical implications for developers.

### 4.3 Interpretation

This empirical study investigated the intuitiveness of bitwise operators through task completion time and error rates. The results highlight that operators can be one of the factors predicting response time. Based on multiple linear regression analysis, we observe a small, negative relation indicating that participants took longer to give a correct answer compared to giving an incorrect one.